

Full Report

"my project name is ai blood grop detection using fingerprint i am bhoomika ng 7th sem ise and its my won project give papper" *Generated by LeapSpace*

AI-Based Blood Group Detection Using Fingerprint Biometrics: Quick Insights

Key Findings Table

Key finding	Population / data scope	Practical meaning	Citations
Fingerprint-based blood group detection is mainly proposed as a non-invasive screening tool, not a replacement for lab typing	Multiple fingerprint blood-group studies and reviews	Best framed for preliminary triage, emergency support, or resource-limited screening	[1] [2] [3] [4] [5] [6] [7]
Deep learning dominates this topic, especially CNN and transfer-learning models such as ResNet, VGG16, MobileNetV2, and hybrid systems	Recent AI studies on fingerprint images	Raw-image learning is currently the main technical direction	[3] [4] [5] [8] [9] [10] [11] [12] [13]
Hybrid/ensemble models often report the strongest results	Internal and some external-validation studies	Combining learned image features with feature-based classifiers may improve performance	[3] [5] [12] [13]
Reported accuracy is often high, but validation rigor is uneven	Studies reporting 0.88 to 99.88% accuracy, often on limited datasets	Strong headline accuracy should be interpreted cautiously	[4] [5] [11] [13] [14] [15] [16] [17] [18] [19]
Dataset quality is a major bottleneck: limited demographic reporting, sparse label-verification details, and weak external validation	Public, private, and hybrid fingerprint datasets	Generalizability to real clinical use is still uncertain	[5] [15] [20] [21] [22]
Preprocessing is treated as important, but many papers report it inconsistently	Fingerprint image pipelines using enhancement, normalization, segmentation, augmentation	Reproducibility is limited unless preprocessing is clearly documented	[1] [23] [24] [25] [26] [27] [28]

Key finding	Population / data scope	Practical meaning	Citations
Biological interpretation remains weak: older dermatoglyphic studies show correlation, not deterministic causation	Observational fingerprint-pattern studies	AI models may predict statistical patterns, but mechanism remains unclear	[5] [15] [29] [30] [31] [32]

Direct Answer

For your project paper, the best evidence-supported title is **“AI-Based Blood Group Detection Using Fingerprint Biometrics”**. The literature supports this as a **promising non-invasive screening project** that can use fingerprint images, preprocessing, and AI models such as CNN, ResNet, MobileNetV2, or hybrid CNN+feature systems to predict ABO and Rh classes, but it should be presented as an **exploratory screening system** rather than a clinically approved replacement for serological blood typing [1] [3] [5] [6] [7] [11].

A strong student paper should therefore:

1. define the problem as **preliminary blood-group prediction from fingerprints** [1] [4];
2. explain the processing pipeline: acquisition, preprocessing, feature extraction or raw-image learning, classification, evaluation [1] [2] [24] [26];
3. compare **classical ML vs deep learning** where possible [11] [33] [34];
4. report metrics beyond accuracy if available, especially F1-score and confusion matrix [5] [35];
5. clearly state limitations: small datasets, class imbalance, weak external validation, privacy concerns, and continued need for serology confirmation [6] [18] [19] [36].

Confidence level: High - the supplied evidence consistently supports fingerprint AI as a screening-oriented research topic, with repeated caution against replacing standard blood typing [1] [3] [6] [7].

Study Scope

This report covers published findings summarized in the supplied evidence on **ABO and Rh blood-group prediction from fingerprint biometrics**, with emphasis on image preprocessing, handcrafted and deep features, machine learning/deep learning classifiers, dataset design, validation, clinical feasibility, and ethics [1] [3] [5] [15] [37]. The evidence includes observational dermatoglyphic studies and recent AI-based fingerprint image classification work, mainly from the 2018-2026 period for AI systems, with some older correlation studies used for background context [11] [29] [32] [38].

Assumptions & Limitations

- Many reported results come from **limited or incompletely described datasets**, so cross-study comparison is imperfect [18] [20] [39].

- External or prospective validation is sparse, reducing confidence in real-world generalization [19] [22].
- Some ethical discussion is borrowed from broader biometric literature because blood-group-specific privacy analysis is limited in the retrieved set [36] [40] [41].
- The evidence supports a **research paper/report**, not a fully submission-ready manuscript with complete tables, figures, and references already formatted [meta-analysis direct answer summarized in 1-2,2-1,4-7,5-11].

Suggested Further Research

- Build larger **multicenter, demographically diverse** fingerprint datasets with verified ABO/Rh labels [15] [42].
- Use **subject-wise and external validation** consistently to reduce leakage and overfitting risk [17] [38].
- Compare **raw-image deep models** against **handcrafted ridge/minutiae/SIFT pipelines** on the same dataset [11] [28].
- Report class-wise metrics for all **eight ABO+Rh classes** rather than overall accuracy alone [11] [35].
- Develop privacy-aware workflows with consent, anonymization, and secure biometric storage [36] [41] [44].

1. Introduction and Problem Framing

AI-based blood group detection using fingerprint biometrics aims to predict a person's ABO and Rh type from fingerprint images instead of drawing blood, making it attractive as a non-invasive, quick, and potentially low-cost screening approach [1] [4] [5] [37] [45]. Across the supplied studies, this idea is repeatedly linked to emergency support, donor pre-screening, and use in settings where immediate laboratory testing may be difficult [5] [46] [47].

However, the evidence does **not** support fingerprint prediction as a substitute for conventional serological typing. Reviews and comparative discussions consistently maintain that serology remains the routine or gold-standard method, while fingerprint-based prediction is better understood as a supplementary or exploratory tool [6] [7] [48]. Taken together, these findings suggest that your project should be framed as an **AI-assisted prediction system for preliminary screening**, not as a diagnostic replacement. **Confidence level: High** - this distinction appears consistently across the evidence [1] [3] [6] [7].

2. Foundational Context

2.1 What the project is actually predicting

Most studies target either:

- the four main ABO groups: **A, B, AB, O**, or
- the full **eight-class ABO+Rh problem**: A+, A-, B+, B-, AB+, AB-, O+, O- [11] [20].

The AI model does not measure antigens directly. Instead, it tries to learn statistical patterns from fingerprint structure or texture that may correlate with blood-group labels in the training data [29] [30] [31] [32]. This matters because correlation does not guarantee biological causation. Older dermatoglyphic studies describe loops, whorls, and arches as associated with some blood-group distributions, but they do not claim that fingerprint patterns determine blood type [29] [30] [31] [32].

2.2 Why fingerprints are used

Fingerprints are easy to capture, non-invasive, and already common in biometric systems, making them appealing for AI experimentation in healthcare screening [4] [37] [49] [50]. From a systems perspective, the attraction is practical: no blood handling, lower direct contact, potentially faster acquisition, and compatibility with digital image pipelines [1] [5] [16] [33] [51].

2.3 Core technical pipeline

The typical workflow in the supplied literature is:

1. **Fingerprint acquisition** [1] [38]
2. **Preprocessing/enhancement** such as normalization, resizing, segmentation, binarization, thinning, denoising, or augmentation [26] [27] [38]
3. **Feature extraction or raw-image learning** using handcrafted ridge/minutiae descriptors or CNN-based representation learning [11] [43]
4. **Classification** using SVC, KNN, XGBoost, CNNs, transfer learning, or hybrid ensembles [5] [11] [12] [33] [34]
5. **Evaluation** using accuracy and, less consistently, F1-score, sensitivity, or external testing [5] [11] [35].

3. Fingerprint Features and Image Processing

3.1 Fingerprint acquisition and quality issues

Fingerprint capture quality directly affects classification performance. One supplied study explicitly notes that some samples could not be tested because of poor image quality, excessive noise, or finger movement during capture, and it also reports cleaning fingertips with alcohol before scanning and standardizing image size and resolution [38]. This indicates that blood-group prediction from fingerprints is sensitive to ordinary biometric capture problems such as motion, noise, and inconsistent contact quality [38].

A limitation of the literature is that sensor type, exact capture conditions, and failure analysis are often underreported, which makes replication difficult [15] [20] [38].

3.2 Preprocessing and enhancement

Preprocessing appears repeatedly as a crucial stage. Reported steps include:

- normalization and standardization [1] [25] [27];
- resizing [25] [27];
- histogram equalization [26];
- binarization [26];
- segmentation [26];
- thinning [26];
- Gaussian blur or other noise reduction [27];
- augmentation using rotation, shear, zoom, and flips [38].

Many papers also describe preprocessing and augmentation as necessary for better generalization of CNN models [2] [14] [23] [24] [45]. At the same time, reporting remains inconsistent: one 2025 fingerprint ABO/Rh study using SIFT and XGBoost did not clearly describe enhancement, normalization, segmentation, or augmentation in the provided excerpt [28]. **Confidence level: Medium** - preprocessing importance is well supported, but exact workflows differ and are not always transparent [1] [26] [27] [28].

3.3 Feature extraction strategies

The literature shows two broad strategies.

Handcrafted feature pipelines

These use explicit fingerprint descriptors such as:

- ridge-flow orientation maps [43];
- ridge frequency [43];
- core/delta geometry [43];
- ridge count between core and delta [43];
- minutiae like ridge endings and bifurcations [43];
- SIFT-based local descriptors [28].

These approaches are more interpretable because they map directly to known fingerprint structures [43].

Raw-image deep learning

Recent studies increasingly use CNNs or transfer learning directly on fingerprint images, allowing the model to learn its own representation without manually defining ridge features [9] [11] [52]. In the raw-image comparative study, ResNet-based modeling outperformed several other deep architectures for blood-group prediction [52].

Taken together, the field is shifting from classical dermatoglyphic descriptors toward deep or hybrid representations [1] [10] [43] [53].

4. Machine Learning and Deep Learning Methods

4.1 Classical machine learning pipelines

The supplied evidence shows feature-based classifiers such as SVC, KNN, and XGBoost in this topic area [28] [33] [34]. A 2025 hemotype-determination study used **SIFT + XGBoost** after extracting ridge-flow and minutiae-related spatial features [28]. Another study used **Support Vector Classifier** with fingerprint-pattern correlations, though the supplied excerpt did not include detailed class-wise metrics [33].

Evidence from related fingerprint classification work suggests traditional ML pipelines may underperform transfer-learning CNNs, with SVM reaching 70.2% accuracy while MobileNetV2 achieved 91.72% accuracy and 95.92% precision in a non-blood-group fingerprint task [34]. This is not a direct blood-group benchmark, but it supports the broader shift toward deep learning for fingerprint-image classification [34].

4.2 CNN and transfer learning models

CNN-based architectures are the dominant modeling choice in fingerprint blood-group prediction [2] [3] [4] [5] [11] [14] [45] [51]. Reported models include:

- custom CNNs [4] [8] [14];
- ResNet variants such as ResNet50 and ResNet152 [9] [11];
- VGG16 [11];
- LeNet [11];
- MobileNetV2 [47] [54];
- Vision Transformer and transformer-style hybrids [10] [12].

Comparative findings highlight that **Vision Transformer** was reported as the best-performing architecture on a public fingerprint blood-group dataset, outperforming CNN, MobileNet, and ResNet+RNN baselines, with authors attributing this to attention mechanisms that capture fine fingerprint details [10]. Among more conventional models, **ResNet50** outperformed custom CNN, VGG16, and LeNet in one ABO/Rh prediction study [11]. Another study favored **ResNet152** for eight-class classification [9].

4.3 Hybrid and ensemble approaches

Hybrid systems are a major theme in the stronger-performing papers. Reported examples include:

- **CNN + SIFT + XGBoost** [35];
- stacked XGBoost ensembles [12];
- weighted hybrid models with external validation [5];
- soft-voting ensembles [13].

These approaches often outperform single backbones. Reported results include **0.8959 accuracy** for a stacked XGBoost model, **96.73% external-validation accuracy** for a weighted hybrid model, and **99.88% accuracy** for a soft-voting ensemble [5] [12] [13]. **Confidence level: High** - multiple supplied studies support the trend that hybrid and ensemble methods can improve performance, though not always under equally rigorous validation conditions [3] [5] [12] [13].

5. Dataset Design and Label Quality

5.1 Public, private, and hybrid datasets

The evidence mentions several dataset designs:

- a **6,000-image** fingerprint dataset with eight blood-group classes [15] [20];
- a **6,000-image public dataset plus 300 external validation samples** [5];
- a **300-sample real-world scanner-collected dataset** with agglutination-confirmed blood groups and subject-wise splitting [38].

One study describes the dataset as coming from the Kaggle Fingerprint Classification collection with ABO+Rh labels across eight classes, but the excerpt did not include serology verification or demographics [20]. By contrast, the 2026 hybrid deep-learning study explicitly confirmed blood groups by agglutination, which strengthens label credibility [38].

5.2 Sample size and class structure

The most repeatedly cited AI datasets are around **6,000 fingerprint images**, generally organized into the eight ABO+Rh classes [5] [15] [20]. Real-world external-style validation appears in a **300-sample** scanner-collected study [38]. Observational non-AI dermatoglyphic work also includes much larger participant cohorts, ranging from hundreds to tens of thousands, but these studies mainly examine statistical association rather than image-based AI prediction [29] [31] [32] [55] [56] [57] [58].

5.3 Label verification and diversity

Label quality varies. The strongest supplied evidence for verified labels is the scanner-collected study using **agglutination-confirmed** blood groups [38]. Some broader summaries mention **clinically verified labels**, but paper-level verification details are not always visible in the excerpts [15]. Demographic reporting is also weak: one dataset is described as diverse, but exact age, sex, and ethnicity distributions are usually absent [20] [25]. This directly limits claims of fairness and generalizability [15].

Confidence level: Medium - dataset concerns are clearly supported, but reporting gaps in the retrieved evidence prevent precise cross-study comparison [15] [20] [38].

6. Model Performance and Validation

6.1 Reported accuracy and class-wise performance

Reported fingerprint blood-group prediction performance spans a wide range:

- about **0.88 prediction accuracy** on eight classes [11];
- **93.14% validation accuracy** for a CNN [4];
- **91.22% training accuracy** in another CNN study [14];
- **96.73% external-validation accuracy** for a weighted hybrid model [5];
- **99.88% accuracy** for a soft-voting ensemble [13].

One hybrid CNN+SIFT+XGBoost study reported **91% overall accuracy**, with class-wise F1 scores ranging from **0.89 to 0.96**, where **B-** achieved the highest F1 and **AB+** the lowest in the provided summary [59]. This shows why class-wise reporting matters: overall accuracy can hide differences between blood-group classes [35].

6.2 Internal, external, and subject-wise validation

Validation quality differs substantially across papers. The strongest validation detail in the supplied set is the **subject-wise split** used in a 300-sample real-world study, preventing overlap of individuals between training and testing [38]. Another key result is the use of **300 real-world external samples**, where a weighted hybrid model achieved **96.73% external-validation accuracy** [5].

However, another supplied source explicitly states that earlier fingerprint blood-group experiments often used small datasets, training-only results, and lacked strict independent external testing [22]. This conflict is important: some newer work reports stronger validation, but the overall literature base still appears validation-limited [5] [22].

6.3 Validation limitations

The evidence repeatedly identifies several threats to reliability:

- **overfitting**, especially with small datasets and large feature/model complexity [18] [60];
- **data leakage** if training and validation are not fully separated [17];
- **class imbalance**, which can bias reported performance [18] [61] [62];
- lack of **multicenter**, **multi-device**, or geographically diverse testing [18] [19] [42];
- fingerprint-specific limits from **small datasets** and insufficient diversity [39] [63].

Confidence level: High - validation weakness is one of the most consistent patterns in the evidence [17] [18] [19] [22].

7. Clinical Feasibility and Real-World Use

The practical appeal of this technology is clear: it could support quick, non-invasive prediction in emergencies, donation camps, and resource-limited settings [5] [46] [47]. Several papers position it as inexpensive, scalable, and suitable for contactless or low-contact workflows [1] [4] [5] [37] [49] [50].

But the evidence still favors a cautious conclusion. Conventional serological typing remains the routine standard, while molecular/genotyping approaches have much stronger reliability evidence for actual blood-group determination [6] [7] [48]. One CRISPR/Cas13a ABO genotyping assay achieved **100% concordance** with serology and sequencing using buccal swabs, illustrating that low-invasive alternatives with direct biological grounding may currently be more practice-ready than fingerprint prediction [7].

Therefore, the most supportable deployment claim is this: **fingerprint AI may be useful as a pre-screening or decision-support layer, but not as a standalone transfusion decision tool** [5] [6] [7].

Confidence level: High - the sources consistently favor this conservative interpretation [1] [4] [6] [7].

8. Ethics, Privacy, and Future Work

8.1 Ethical and privacy concerns

Blood-group-specific ethics discussions are limited in the supplied studies, but broader biometric literature raises clear concerns relevant to this project. Fingerprints are sensitive personal identifiers, difficult to revoke if leaked, and potentially vulnerable to misuse, surveillance, cross-system identification, and discrimination [36] [41] [64] [65] [66]. Several studies recommend informed consent, anonymization, secure storage, access control, and limiting use to research purposes [36] [44] [67].

For a student project, a privacy-aware design should explicitly state:

- consent before collecting fingerprints [36];
- secure and minimal data storage [44];
- no secondary reuse without permission [67];
- prediction used only for research or screening, not treatment decisions [6] [7].

8.2 Future research directions

The strongest evidence-backed future directions are:

1. larger, multicenter datasets with demographic diversity [15] [42];
2. clinically verified labels through serology/agglutination [15] [38];
3. stronger subject-wise, external, and prospective validation [17] [38];
4. direct comparison of handcrafted and deep representations [11] [28];
5. better reproducibility through clear preprocessing and ablation studies [68] [69] [70];
6. explicit privacy governance for biometric-health inference [36] [41].

9. Project-Oriented Conclusion

For Bhoomika NG's 7th semester ISE project, **AI-based blood group detection using fingerprint** is a valid and timely research topic if it is positioned correctly. The literature supports it as a **non-invasive AI screening concept** with promising results from CNNs, ResNet-family models, Vision Transformer approaches, and hybrid ensembles [5] [10] [11] [13]. Preprocessing and image quality control are important, and feature strategies range from ridge/minutiae descriptors to raw-image deep learning [26] [43].

At the same time, the field remains **exploratory** because of inconsistent preprocessing reporting, incomplete dataset transparency, weak demographic documentation, and insufficient independent real-world validation [15] [18] [20] [22]. Conventional serology remains the gold standard, so the safest and most academically strong framing is: **“an AI-based fingerprint system for preliminary ABO/Rh prediction and screening support, with confirmation by standard blood typing.”** [6] [7]

Overall conclusion confidence: High - this conclusion aligns with the dominant pattern across technical, validation, and clinical-feasibility evidence [1] [3] [4] [5] [6] [7].

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